Venkata Abhinav Kumar Gupta Gunda

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LinkedIn | **GitHub**

Education

Vellore Institute of Technology — **Chennai, TN**
B.Tech in Computer Science and Engineering; CGPA: 7.95 — *Aug 2023 – Present*

Turito Junior College — **Hyderabad, TS**
Intermediate (M.P.C) — Board Score: 92.7% — *Jun 2021 – Apr 2023*

Narayana Olympiad School — **Proddatur, AP**
Secondary School Certificate (SSC) — GPA: 9.9/10 — *Jul 2020 – May 2021*

Experience

VIC IT Consulting — **Johannesburg, South Africa (Remote)**
Security Engineer Intern

- Executed **Vulnerability Assessments and Penetration Testing (VA/PT)** across web applications, network infrastructures, and cloud environments using **Burp Suite**, **Nmap**, and **Acunetix**.
- Performed targeted reconnaissance in **Kali Linux** to identify **OWASP Top 10** vulnerabilities (SQLi, XSS, Broken Authentication), drafting enterprise remediation reports.
- Assisted in real-time **threat evaluation and attack vector analysis** to harden client infrastructure and streamline internal **SecOps** testing lifecycles.

Key Projects

RAGShield: Automated RAG Security Assessment Framework | *FastAPI, ChromaDB, Ollama* — **May 2026 – Aug 2026**

- **Architected an end-to-end security scanner** across 4 RAG layers: document ingestion, vector storage (**ChromaDB**), semantic retrieval, and LLM inference (**Ollama**).
- **Engineered adversarial testing modules** simulating **document poisoning**, **indirect prompt injection (IPI)**, vector ranking manipulation, and data leakage.
- **Developed an analysis engine** in **FastAPI** correlating similarity score drifts and access violations into automated severity and confidence scoring with **PoC traces**.

Surgical-Guard: Semantic Middleware for AI Agent Security | *Transformers.js, Chrome API, WASM* **Jan 2026 – Mar 2026**

- Architected a **Browser-Based AI Firewall** to mitigate **Indirect Prompt Injection (IPI)** and **data exfiltration** in autonomous LLM-integrated workflows.
- Engineered a **Local-First Semantic Engine** using **Transformers.js** for real-time intent analysis, intercepting paraphrased attacks and role-shift hijacking (<100ms latency).
- Implemented **Instruction-to-Data (I2D) Transmutation**, safely neutralizing hostile instructions into passive data while preserving **92% system utility**.

Agentic Multimodal Threat Orchestrator | *Gemini, Random Forest, OpenCV, ChromaDB* — **Oct 2025 – Dec 2025**

- Architected a **Decentralized Multi-Agent System** using **Gemini** and **Random Forest** for cross-modal correlation between network telemetry and surveillance feeds.
- Engineered **Perception Agents** identifying cyber-physical attack chains (e.g., brute-force logins paired with unauthorized facility entry), achieving a **42% reduction in FPR**.
- Deployed an **Explainable AI (XAI) Pipeline** in Flask providing verifiable **Chain-of-Thought** logs for automated **SOC auditability** reports.

Technical Skills

- **Programming:** C, C++, Java, Python, JavaScript
- **Web Development:** HTML, CSS, ReactJS, WebAssembly, Chrome Extensions
- **Core Competencies:** AI Security, LLM Red Teaming, Network Security, Penetration Testing
- **Tools & Platforms:** Git, Firebase, Splunk, Wireshark, Nmap, ChromaDB, Transformers.js
- **Databases:** MySQL, Firebase Firestore (NoSQL)
- **Languages:** Telugu (Native), English (Fluent), Hindi (Intermediate), German (Basic)

Achievements & Certifications

- **Cybersecurity & Ethical Hacking Certification** – iHUB DivyaSampark, IIT Roorkee in association with Intellipaat. View Certificate
- **Adobe Hackathon:** Developed an offline-first document intelligence system for automated hierarchy extraction and PDF QA.
- **IBM Cybersecurity Analyst Certification** – Verified professional credential via IBM SkillsBuild. View Certificate
- **Cybersecurity Job Simulations** – Completed Deloitte Australia & Tata analyst tracks via Forage (Aug 2025).
- **AWS Cloud Technology Training** – Sri Sathya Sai Seva Organizations (Jul – Aug 2025). View Certificate